Hotel Booking Data Analysis Report

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1. Introduction

This analysis explores booking data from an online hotel reservation platform to identify booking trends, customer behavior, cancellation patterns, and revenue opportunities. The objective of this analysis is to derive actionable insights that can help improve operational efficiency, reduce cancellations, and optimize revenue strategies.

The dataset contains booking records including information about customers, hotel properties, cities, star ratings, booking channels, booking values, payment methods, and booking status. Using exploratory data analysis and visualization techniques, key patterns were identified to support business decision-making.

1.1 Dataset Summary:

The dataset used in this analysis contains booking information from an online hotel reservation platform. Each record in the dataset represents a single hotel booking made by a customer.

The dataset includes **30,000 booking records** and **24 different columns**, covering details about customers, hotels, booking channels, pricing, and booking outcomes. This information helps us understand how customers book hotels, how much they spend, and what factors influence cancellations or successful bookings.

The data includes several types of information such as:

- **Customer and hotel details** – including customer ID, hotel property ID, city, and hotel star rating.
- **Booking information** – such as booking date, check-in date, check-out date, number of rooms booked, and type of stay.
- **Booking channel details** – which show whether the booking was made through the website, mobile app, or travel agent.
- **Pricing information** – including booking value, cost price, selling price, markup, cashback, and coupon usage.
- **Booking outcomes** – which indicate whether the booking was confirmed, cancelled, or failed, along with any refund information.

Before starting the analysis, the dataset was cleaned and prepared. Date columns were converted into proper date formats so that time-based trends such as monthly bookings and stay duration could be analyzed. New fields such as **stay length** and **booking month** were also created to better understand customer travel behavior.

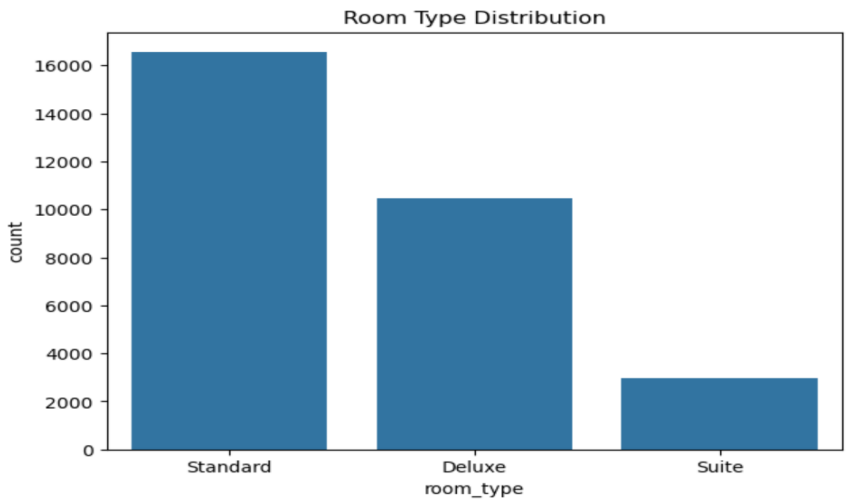
This prepared dataset allowed us to explore booking patterns, identify cancellation trends, and generate insights that can help improve the platform's performance.

2. Key Observations

2.1 Meaningful Trends and Patterns:

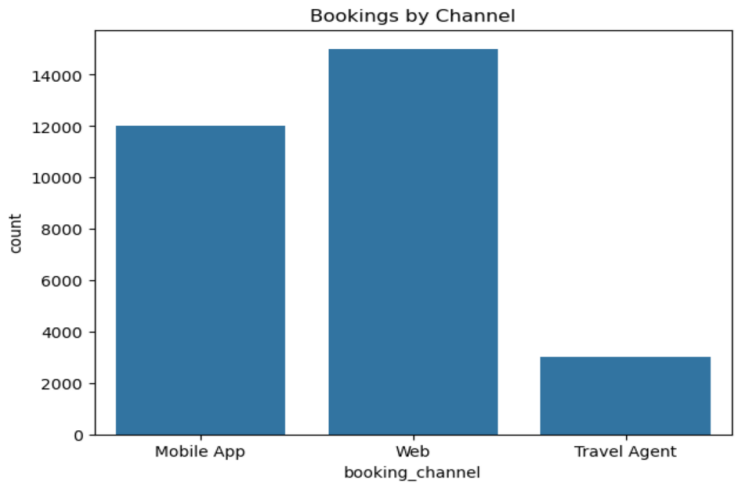
1. Most Customers Prefer Budget-Friendly Options

The analysis shows that **Standard rooms receive the highest number of bookings** compared to Deluxe or Suite rooms. This suggests that most customers prefer **affordable accommodation options** rather than premium hotel rooms. While luxury rooms generate higher booking values, they are booked less frequently.



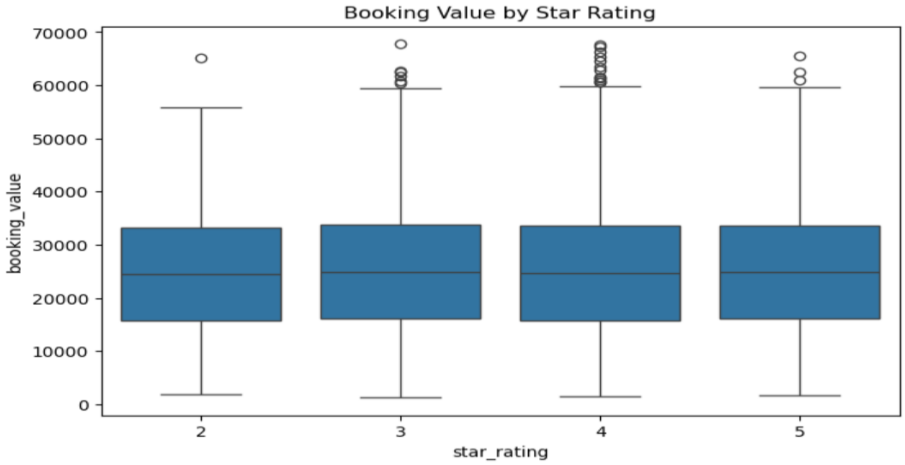
2. Website is the Most Popular Booking Channel

Among the different booking channels, the **Web channel accounts for the highest number of bookings**. This indicates that customers may find the website easier to use or more reliable when making reservations. Mobile app bookings are slightly lower, which may suggest that customers still prefer booking through a desktop interface when making travel plans.



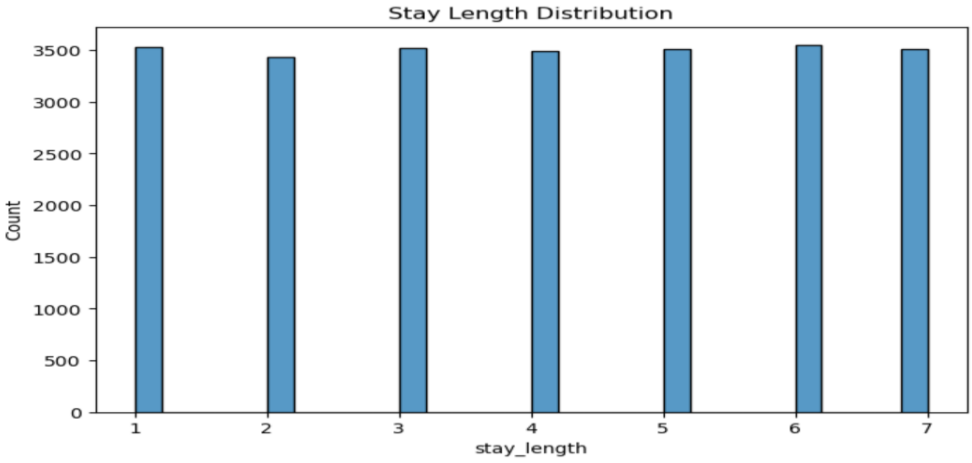
3. Higher Star Hotels Generate Higher Booking Values

Hotels with higher star ratings generally show **higher booking values**. Customers who choose 4-star or 5-star hotels tend to spend more per booking, which indicates that hotel quality and amenities strongly influence customer spending behavior.



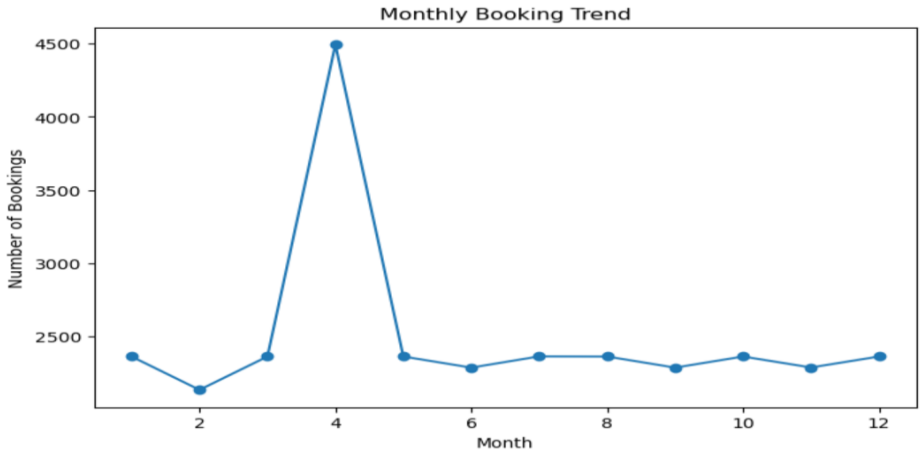
4. Short Stays Are More Common Than Long Vacations

The stay length analysis indicates that most customers book **short-duration stays** rather than long vacations. This suggests that many bookings may be related to **weekend travel, short leisure trips, or business visits**.



5. Monthly Booking Trend

The monthly booking trend shows that the number of bookings remains relatively consistent throughout most months of the year. However, there is a noticeable increase in bookings during **April**, which indicates a peak in travel activity during this period.



This spike may be influenced by several factors such as **holiday seasons, school vacations, or increased tourism during that time of the year**. Apart from this peak, booking activity remains fairly stable across the remaining months, suggesting that the platform experiences steady demand throughout the year.

6. Cancellation Trend

The analysis of booking status shows that a noticeable portion of bookings are cancelled before the stay actually takes place. **The cancellation rate is approximately 20% of the total bookings, which means that about one out of every five bookings does not result in a completed stay.**

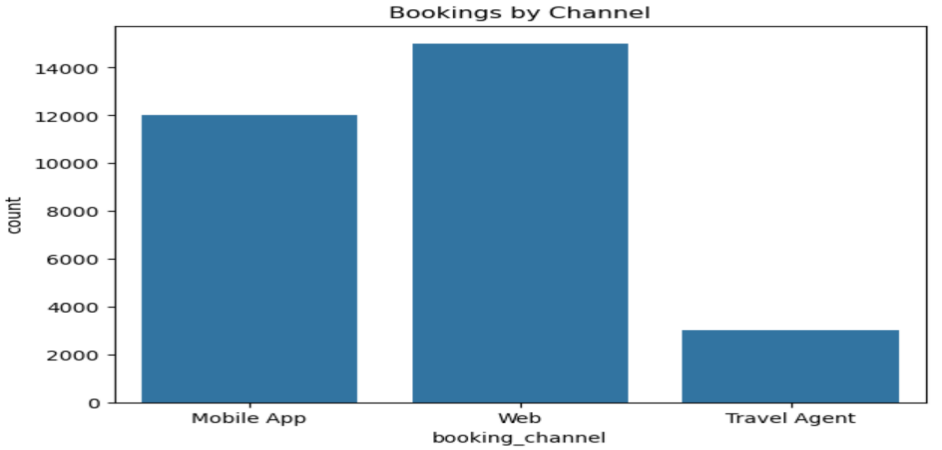
This level of cancellations suggests that many customers may be making flexible or tentative reservations while still finalizing their travel plans. In some cases, customers may book multiple options and later cancel the ones they do not need. Another possible reason could be changes in travel schedules, price comparisons with other platforms, or attractive last-minute offers elsewhere.

Booking Status	**Cancelled**	**Confirmed**	**Failed**
Booking Channel			
Mobile App	**2589**	**8111**	**1309**
Travel Agent	**835**	**1979**	**176**
Web	**2646**	**11582**	**773**

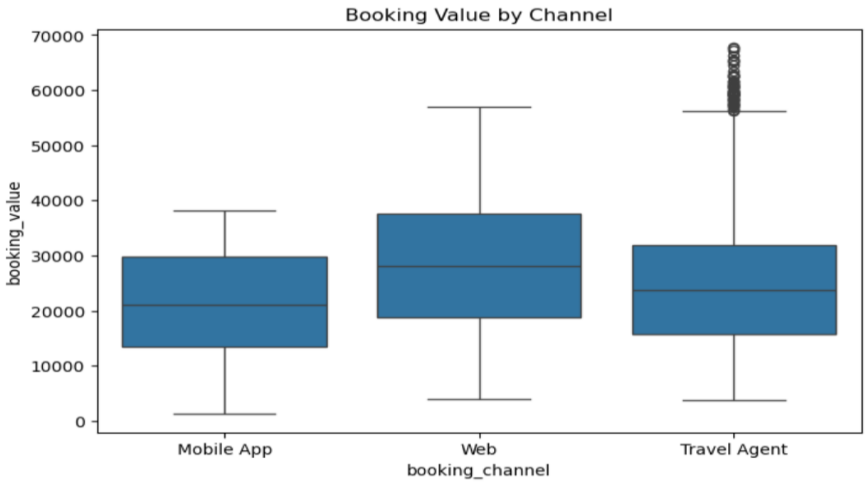
The data shows that cancellations occur across all booking channels, but they are slightly higher in the **Web and Mobile App channels** because these platforms have the largest number of bookings. In comparison, the **Travel Agent channel has fewer cancellations**, which suggests that bookings made through agents are usually more finalized. Overall, about **20% of the bookings are cancelled**, meaning roughly **1 out of every 5 bookings does not result in a completed stay**.

2.2 Booking Patterns Across Channels, Room Types, and Star Ratings

Booking Channels:
The Web channel generates the highest number of bookings compared to Mobile App and Travel Agent channels. This indicates that customers may prefer booking directly through the website, possibly due to easier navigation, better offers, or familiarity with the platform.



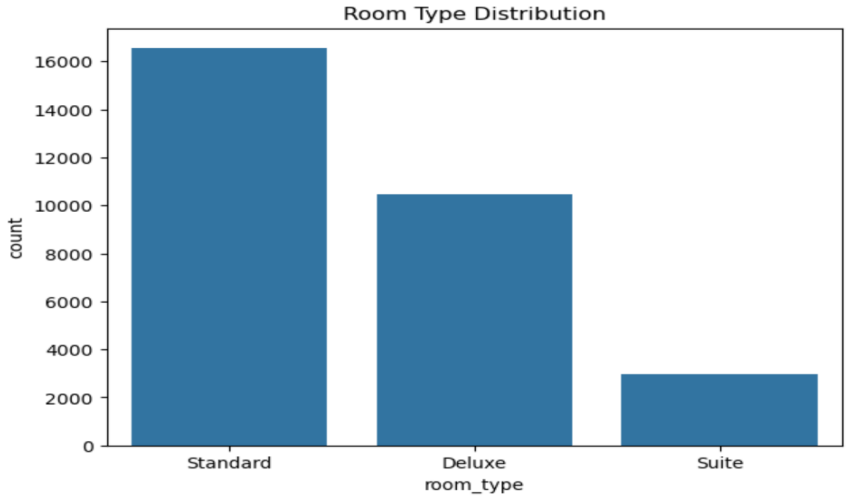
During the analysis of booking values across different booking channels, it was observed that the **Travel Agent channel contains several high-value outliers compared to Web and Mobile App bookings**. This means that while the total number of bookings through travel agents is lower, some of these bookings involve significantly higher booking amounts.



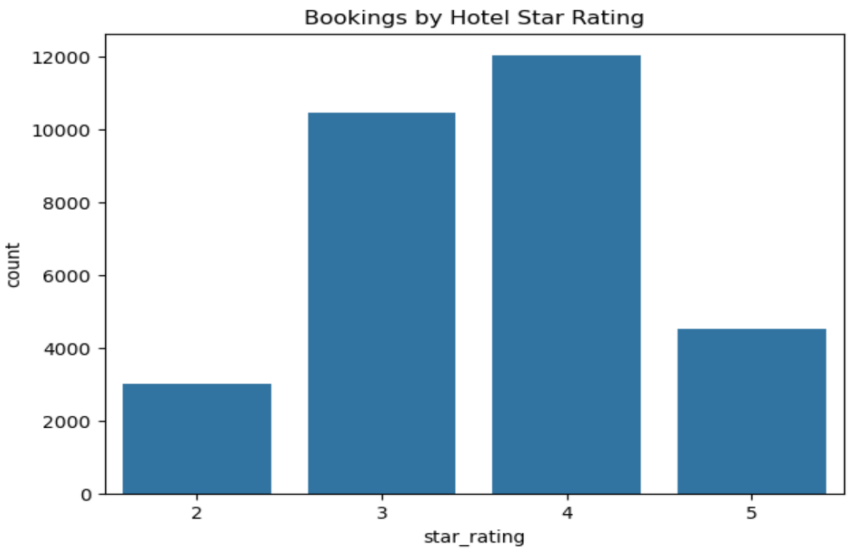
This pattern suggests that travel agents may handle **larger or premium bookings**, such as group reservations, luxury hotel stays, or corporate travel arrangements. Unlike individual customers booking online, travel agents often manage bookings for multiple travelers or higher-end accommodations.

As a result, even though travel agents contribute fewer bookings overall, they may still play an important role in generating **high-value transactions for the platform**.

Room Type Preferences:
Standard rooms account for the majority of bookings, suggesting that customers tend to prefer more affordable accommodation options. Premium room types such as Deluxe and Suite rooms receive fewer bookings but may generate higher booking values.

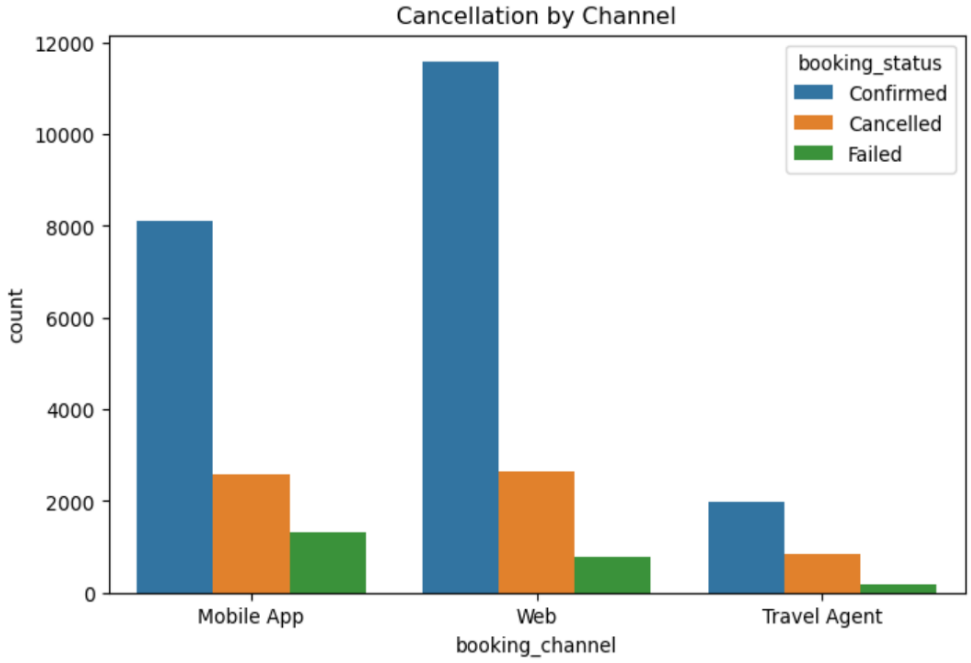


Hotel Star Ratings
Higher star-rated hotels generally show higher booking values. Customers choosing luxury hotels are likely willing to spend more for better amenities and comfort.



2.3 Cancellation Trend Analysis

The analysis shows that approximately **20% of bookings are cancelled**, indicating that roughly one in five reservations does not convert into an actual stay.



The chart shows the number of **confirmed, cancelled, and failed bookings across different booking channels** such as Mobile App, Web, and Travel Agent.

From the chart, we can see that the **Web channel has the highest number of confirmed bookings**, which means most customers prefer booking hotels through the website. The Mobile App also contributes a significant number of bookings, but it is slightly lower than the Web channel.

In terms of cancellations, both the **Web and Mobile App channels show a noticeable number of cancelled bookings**, which is expected because these channels have the largest number of total bookings. On the other hand, the **Travel Agent channel has the lowest number of bookings and cancellations**, indicating that bookings made through agents are usually more planned and less likely to be cancelled.

Overall, the diagram suggests that **online platforms (Web and Mobile App) drive most of the bookings**, while **travel agent bookings are fewer but relatively more stable with fewer failures and cancellations**.

3. Root Cause Analysis

3.1 Reasons for Cancellation Patterns

The analysis shows that a noticeable number of bookings are cancelled. There can be several reasons behind this pattern. One possible reason is the availability of **flexible cancellation policies**, which allow customers to cancel their bookings **without paying any penalty**. Because of this flexibility, many customers feel comfortable cancelling their reservations if their plans change.

Another reason could be that **customers often book multiple hotels at the same time while comparing prices and options. After finalizing their decision, they cancel the remaining bookings**. In addition, last-minute changes in travel plans, such as schedule adjustments or personal reasons, may also lead to cancellations. Some users may also temporarily book hotels while they are still deciding on their travel plans.

3.2 Why Some Channels Perform Better

The analysis also shows that certain booking channels perform better than others. One reason could be that **website bookings provide a smoother and more reliable user experience**, making customers more comfortable completing their reservations.

Direct bookings through the website may also offer better pricing transparency, allowing users to clearly compare prices and available options. Travel agent bookings, although fewer in number, may involve higher-value reservations because **customers often rely on agents for more organized or premium travel plans.** On the other hand, mobile app bookings may attract users who prefer quick and convenient booking options, especially those who are more price-sensitive.

3.3 Seasonal or Temporal Trends

Seasonal and temporal trends appear to have a **noticeable influence on booking values**. During certain months of the year, **travel demand tends to increase, which can lead to a higher number of bookings and potentially higher booking values.** For example, periods such as holiday seasons, school vacations, and peak tourism months often encourage more people to travel.

When demand increases during these periods, customers may be more willing to spend on accommodation, resulting in higher overall booking values. **Hotels may also adjust their pricing during peak seasons, which can further increase the average booking amount.**

On the other hand, during off-season months **when travel demand is lower, booking values may decrease as fewer people plan trips and hotels may reduce prices to attract customers.** Therefore, these seasonal patterns play an important role in influencing both the number of bookings and the value of those bookings over time.

4. Business Recommendations

4.1 Strategies to Reduce Cancellations

To reduce the cancellation rate, several practical strategies can be adopted:

• Introducing **partial cancellation fees for last-minute cancellations** may discourage users from cancelling bookings unnecessarily.

• The platform can **encourage confirmed bookings by offering incentives**, such as discounts on future bookings or loyalty reward points.

• Sending **reminder notifications before the check-in date** can help customers remember their reservations and reduce accidental or last-minute cancellations.

• During **peak travel seasons**, the platform may implement slightly **stricter cancellation policies** to ensure more stable and reliable bookings.

4.2 Improve Profitability and Increase Repeat Bookings

The platform can improve both revenue generation and customer retention by implementing the following strategies:

• Introducing **loyalty programs** can encourage customers to make repeat bookings by rewarding them with points, discounts, or exclusive benefits.

• Providing **personalized hotel recommendations based on previous booking history** can improve the customer experience and increase the likelihood of future bookings.

• The platform can also **promote premium hotel options to customers who show higher spending behaviour**, which can help increase the overall booking value.

• Another strategy is to **bundle hotel bookings with additional travel services**, such as transportation, tours, or local activities, creating more value for customers while increasing total revenue.

4.3 Optimize Pricing, Promotions, and Channel Strategy

Based on the analysis, the platform has several opportunities to improve its business strategies:

• The company can **focus its marketing efforts on the booking channels that generate the highest number of reservations**, as these channels already attract more customers.

• During **months with lower demand**, the platform can introduce targeted promotions or special offers to encourage more bookings and maintain steady revenue.

• Providing **special discounts or promotional offers for high-demand cities** can help attract more customers and increase the platform's market share in popular travel destinations.

• The platform can also **apply dynamic pricing strategies during peak travel periods**, adjusting prices based on demand to maximize revenue while still remaining competitive.